

Project Design Phase

Problem – Solution Fit Template

Date	28 June 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) Small & medium-scale farmers Agricultural researchers & scientists Govt. policymakers & agri-stakeholders Rural farming communities across India	CS	6. CUSTOMER CONSTRAINTS Limited digital literacy among rural farmers No internet in remote areas Low budget for agri-tech tools Lack of soil testing equipment Language barriers (regional)	CC	5. AVAILABLE SOLUTIONS Manual soil testing labs (slow, expensive) Traditional farming knowledge Generic govt. crop guidelines Costly agri-tech apps (limited rural access) Extension officer advice	AS
	2. JOBS-TO-BE-DONE / PROBLEMS Select the right crop for local soil & climate conditions Avoid fertilizer waste (N,P,K) Maximize yield & income Assess crop suitability before planting season Plan farming strategy with data	J&P	9. PROBLEM ROOT CAUSE No affordable, data-driven tool for individual farm-level crop selection in India Farmers lack access to soil analytics + crop advisory in one integrated platform	RC	7. BEHAVIOUR Farmers rely on neighbors or tradition for crop selection Trial-and-error planting cycles Visit local agri officers rarely Researchers use spreadsheets for manual data analysis	BE
Identify strong TR & EM	3. TRIGGERS Poor harvest season Soil test shows nutrient gaps Govt. awareness campaigns Peer farmer success stories	TR	10. YOUR SOLUTION (OptiCrop) ML-powered web app (Flask) trained on agricultural data Input: N, P, K, Temperature, Humidity, pH, Rainfall Output: Best crop recommendation + suitability score Visualization dashboard for researchers (Matplotlib/Seaborn) Simple UI — no tech expertise needed by farmers	SL	8.1. CHANNELS — ONLINE Flask web application (browser) Mobile-friendly interface Agri-portal integrations	CH
	4. EMOTIONS: BEFORE / AFTER Before: Anxious, uncertain about crop choice & yield outcome After: Confident, informed, resource-efficient farming	EM			8.2. CHANNELS — OFFLINE Agri extension officer demos Farmer training workshops Local govt. agri-centers	CH